

Tanay Chitlur

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[Google Scholar](#) | [Portfolio: papers, posters & presentations](#)

Summary

Carnegie Mellon computer science student and published machine learning researcher. First author on two peer-reviewed IEEE papers (EMBC, Sensors) applying transformer architectures to biomedical signal synthesis. Two-time ISEF Grand Award winner and 2026 Coca-Cola Scholar.

Education

Carnegie Mellon University <i>Bachelor of Science in Computer Science</i> • Relevant Coursework: Mathematical Foundations of Computer Science (Discrete Math), Principles of Imperative Computation, Matrices and Linear Transformations, Calculus 1–3	Pittsburgh, PA Aug. 2026 – May 2030
Westview High School <i>High School Diploma — GPA: 3.98 Unweighted / 4.46 Weighted</i>	Portland, OR Graduated May 2026

Research and Publications

Transformer-Based Framework with Dynamic Multiscale Attention Mechanisms for 12-Lead ECG Synthesis • First-authored and presented a peer-reviewed paper at the 47th IEEE EMBC, the largest biomedical engineering conference; indexed in IEEE Xplore and PubMed (Link)	2025
A Morphologically-Aware Transformer Network for High-Fidelity V1–V5 ECG Lead Generation • First-authored and presented a peer-reviewed paper at the 24th IEEE Conference on Sensors; indexed in IEEE Xplore (Link)	2025
Validation of Einthoven and Goldberger’s Electrocardiographic Lead Systems • Published and presented validation research at the Heart Rhythm Society Annual Meeting (HRS’24) (Link)	2024
Synthesis of a Diagnostic-Caliber 12-Lead Electrocardiogram Using a Novel Multi-resolution Transformer • Built a multi-resolution transformer for 12-lead ECG synthesis; won an ISEF Grand Award (2nd place) and presented at HRX 2024 (Link)	2024
IntelliCane: A Deep Learning Cane for Active Obstacle Avoidance & Navigation for the Visually Impaired • Designed a deep learning navigation system for real-time obstacle avoidance; won an ISEF Grand Award (3rd place) and the AAAI Special Award (Link)	2025
Early Detection and Synthesis of a 12-Lead ECG Using Density-Based Clustering and GANs • Presented generative modeling research at AI Med 2023, the youngest of 80+ pre- and post-doctoral presenters (Link)	2023

Work Experience

Machine Learning Engineering Intern <i>Hello Alfred.ai</i> • Built an ML-driven medical history bot for a 24/7 AI health companion using [[ADD TECH STACK]], triaging patient history to flag when clinical care is needed • Presented the system at the 2025 Baylor Scott & White Tech4Better Symposium	2024 – 2025 <i>Remote</i>
Founder <i>TnT Treats® Baking LLC</i> • Founded and scaled an artisanal dessert business to \$15K+ in lifetime sales (Website)	2019 – Present <i>Portland, OR</i>
Team Captain & Software Lead <i>FIRST Tech Challenge — Team 18108, High Voltage</i> • Wrote autonomous control software [[ADD ALGORITHM / CV DETAIL]] and led hardware, software, and outreach for a 4-year competitive robotics team [[ADD COMPETITION RESULT]]	2021 – 2025 <i>Portland, OR</i>

Awards

Coca-Cola Scholar, 2026 — \$20,000 merit-based national scholarship; 1 of 150 selected from 110,000+ applicants

Technical Skills

Languages: Python, Java, C, C++, Swift, JavaScript, HTML, CUDA
ML & AI: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face, XGBoost
Data & Tools: Pandas, NumPy, SciPy, Matplotlib, Seaborn, h5py, React, Linux, AWS, Cursor, Claude Code, GPT Codex, Fusion 360